ONE-PAGE BRIEF

Post-quantum cryptography review: libwally-core @ 374df231d054

Client: Blockstream (libwally) — public repository (passive review) · Source: github.com/ElementsProject/libwally-core, commit `374df231d054` · Reviewed: 2026-11-04 18:56 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 0 of 4

no post-quantum cryptography

FINDINGS

0 Critical **0** High **7** Medium

0 Low **1** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium	QSI-2026-12082	**Quantum-vulnerable RSA key (4096-bit) in SECURITY.md** What to do: Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it..
Medium	QSI-2026-12085	**Quantum-vulnerable ECDH: key wrapping and data encryption at rest (2 file(s))** confirmed by the analyst What to do: Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may already have been copied needs its...
Medium	QSI-2026-12087	**Quantum-vulnerable ECDSA: transaction and message signing (37 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12088	**Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (7 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...

3 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium key wrapping and data encryption at rest	ECDH	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may...
Medium transaction and message signing	ECDSA, Schnorr (BIP-340)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	ECDH, ECDSA, Schnorr (BIP-340)	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Analyst review: 6 confirmed, 1 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-29)

Static review of source code and manifests; files analysed: 249. No requests were sent to the owner's systems.

Full report, content SHA-256: `c5113cb67ef52870716f434261cef10b8348209e2f0bfb7ed78e283dc233aa13`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.